Mathematics for Computing

Assignment 01
Sets and Relations

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Sets

Definition

A **set** is a well-defined, unordered collection of distinct objects.

Mathematical Representation

$$A = \{x \mid P(x)\}$$

where $P(x)$ is a predicate that determines whether x belongs to the set.

Example

$$F = \{\text{Age}, \text{Income}, \text{Experience}, \text{Education}\}$$

represents the feature set used to train a machine learning model.

A set can be represented in two standard forms.

Roster Form
Lists all elements of the set.

$\{1, 2, 3, 4, 5\}$

Set Builder Form
Defines the set using a common property.

$\{x \in \mathbb{N} \mid 1 \leq x \leq 5\}$

$$\boxed{\{1, 2, 3, 4, 5\} = \{x \in \mathbb{N} \mid 1 \leq x \leq 5\}}$$

Set Builder Form

Definition

A set-builder form defines a set by specifying the property satisfied by its elements.

$$A = \{x \mid P(x)\}$$

where $P(x)$ is a predicate.

Instead of listing every element, we describe the mathematical rule that generates the set or satisfies a condition.

1. $\{1, 2, 3, 4, 5, 6, 7\} = \{x \in \mathbb{N} \mid 1 \leq x \leq 7\}$

2. $\{5, 10, 15, 20, 25\} = \{5n \mid 1 \leq n \leq 5,\ n \in \mathbb{N}\}$

3. $\{3, 6, 9, 12, 15\} = \{3n \mid 1 \leq n \leq 5,\ n \in \mathbb{N}\}$

4. $\{1, 4, 9, 16, 25, 36\} = \{n^2 \mid 1 \leq n \leq 6,\ n \in \mathbb{N}\}$

5. $\{0, 2, 4, 6, 8\} = \{x \in \mathbb{N} \mid 0 \leq x \leq 8,\ 2 \mid x\}$

Assignment Questions (6–10)

6. $\left\{\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \frac{5}{6}\right\} = \left\{\frac{n}{n+1} \mid n \in \mathbb{N},\ 1 \leq n \leq 5\right\}$

7. $\{2, 4, 8, 16, 32\} = \{2^n \mid n \in \mathbb{N},\ 1 \leq n \leq 5\}$

8. $\{2, 6, 18, 54, 162\} = \{2 \cdot 3^n \mid n \in \mathbb{N},\ 0 \leq n \leq 4\}$

9. $\{3, 8, 15, 24, 35, 48\} = \{n^2 - 1 \mid n \in \mathbb{N},\ 2 \leq n \leq 7\}$

10. $\{2, 3, 5, 7, 11, 13, 17\} = \{x \in \mathbb{N} \mid x \text{ is prime},\ x \leq 17\}$

Types of Sets

Cardinality

$$|A| = \text{Number of elements in } A$$

Example:

$$A = \{2, 4, 6, 8\}$$

$$|A| = 4$$

Finite Set

$$|A| = n, \qquad n \in \mathbb{N}$$

Example:

$$Classes = \{\text{Cat}, \text{Dog}, \text{Bird}\}$$

$$|Classes| = 3$$

Infinite Set

$$|A| = \infty$$

Example:

$$\mathbb{N} = \{1, 2, 3, \ldots\}$$

Null (Empty) Set

$$A = \varnothing \qquad \text{or} \qquad |A| = 0$$

Example:

$$FP = \varnothing$$

Singleton Set

$$|A| = 1$$

Example:

$$W = \{w^*\}$$

where w^* is the unique optimal weight vector.

Equal Sets

$$A = B \iff (\forall x)(x \in A \Leftrightarrow x \in B)$$

Example:

$$\{1, 2, 3\} = \{3, 2, 1\}$$

Equivalent Sets

$$|A| = |B|$$

Example:

$$|Training| = 1000$$
$$|Validation| = 1000$$

Subset

$$A \subseteq B$$

Example:

$$Dogs \subseteq Animals$$

Superset

$$B \supseteq A$$

Example:

$$Animals \supseteq Dogs$$

Universal Set

$$U = \{\text{All Images}\}$$

Example:

$$Cats,\ Dogs,\ Cars \subseteq U$$

Power Set

$$P(A) = \{X \mid X \subseteq A\}$$

Example:

$$A = \{1, 2\}$$

$$P(A) = \{\varnothing, \{1\}, \{2\}, \{1, 2\}\}$$

Countable Set

$$f : \mathbb{N} \to A \textit{ is a bijection.}$$

Example:

$$\mathbb{N},\ \mathbb{Z},\ \mathbb{Q} \textit{ are countable},\ \textit{while}$$

$$\mathbb{R} \textit{ is uncountable}.$$

Set Operations

Set Operations

Definition

Set operations combine or modify sets to create new sets.

$$A, B \rightarrow \text{New Set}$$

Common set operations:

1. Union
2. Intersection
3. Set Difference
4. Symmetric Difference
5. Complement
6. Cartesian Product

Mathematical Definition

The union of two sets contains all the elements that belong to either set.

$$A \cup B = \{x \mid x \in A \vee x \in B\}$$

Example:

$$A = \{1, 2, 3\}$$
$$B = \{3, 4, 5\}$$
$$A \cup B = \{1, 2, 3, 4, 5\}$$

Example: Combining two feature sets from different datasets.

$$Features = A \cup B$$

Mathematical Definition

The intersection contains elements common to both sets.

$$A \cap B = \{x \mid x \in A \land x \in B\}$$

Example:

$$A = \{Python, Java, C++\}$$

$$B = \{Python, R, Java\}$$

$$A \cap B = \{Python, Java\}$$

Example: Common features selected by two machine learning models.

$$Selected = A \cap B$$

Mathematical Definition

The difference of two sets contains elements present in one set, but not in the other.

$$A - B = \{x \mid x \in A \land x \notin B\}$$

Example:

$$A = \{1, 2, 3, 4, 5\}$$
$$B = \{3, 4, 5, 6, 7\}$$
$$A - B = \{1, 2\}$$

Example: Removing already processed data samples:

$$NewData = Dataset - Processed$$

Mathematical Definition

The symmetric difference contains elements that belong to exactly one of the sets.

$$A \triangle B = (A - B) \cup (B - A)$$

Example:

$$A = \{1, 2, 3\}$$
$$B = \{3, 4, 5\}$$
$$A \triangle B = \{1, 2, 4, 5\}$$

Example: Files modified in either branch but not both.

$$Changes = A \triangle B$$

Mathematical Definition

The complement contains elements from the universal set that are not present in the given set.

$$A^c = U - A$$

Example:

$$U = \{1, 2, 3, 4, 5\}$$
$$A = \{1, 2\}$$
$$A^c = \{3, 4, 5\}$$

Example: Classification errors:

$$Correct^c = Incorrect$$

Mathematical Definition

Cartesian product creates ordered pairs.

$$A \times B = \{(a, b) \mid a \in A, \; b \in B\}$$

Example:

$$A = \{1, 2\}$$

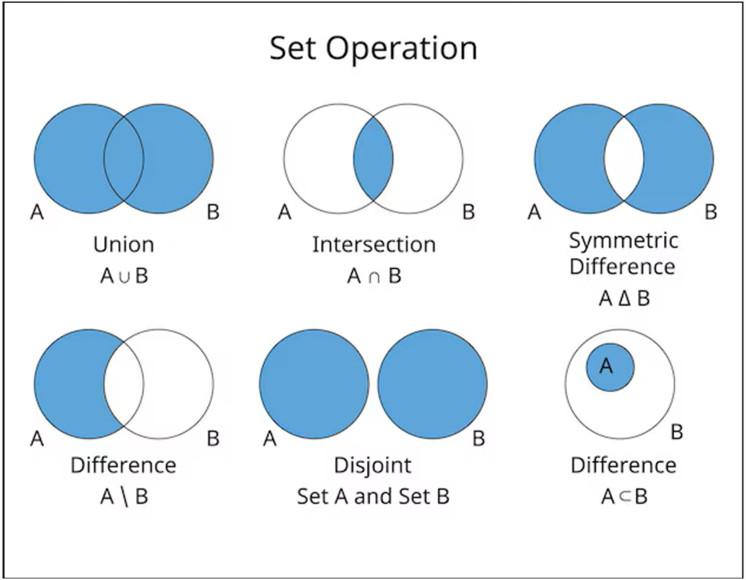
$$B = \{Python, Java\}$$

$$A \times B = \{(1, Python), (1, Java), (2, Python), (2, Java)\}$$

Example: Input-output pairs in supervised learning:

$$(X, Y) \ where$$

$$X \times Y \ represents \ possible \ input - label \ combinations.$$


Set Operation
A
B
Union
A ∪ B
A
B
Intersection
A ∩ B
A
B
Symmetric Difference
A Δ B
A
B
Difference
A \ B
A
B
Disjoint
Set A and Set B
A
B
Difference
A ⊂ B

Inclusion-Exclusion Principle

Definition

The Inclusion-Exclusion Principle is used to determine the cardinality of the union of overlapping sets without counting common elements more than once.

For two finite sets,

$$|A \cup B| = |A| + |B| - |A \cap B|$$

Suppose an autonomous driving model detects

$$A = \{\text{Pixels predicted as Road}\}$$

$$B = \{\text{Ground Truth Road Pixels}\}$$

Suppose

$$|A| = 1200$$

$$|B| = 1100$$

$$|A \cap B| = 950$$

Then

$$|A \cup B| = 1200 + 1100 - 950 = 1350$$

The union represents all pixels that are either predicted as road or actually belong to the road region.

For n finite sets,

$$\left|\bigcup_{i=1}^{n} A_i\right| = \sum |A_i| - \sum |A_i \cap A_j| + \sum |A_i \cap A_j \cap A_k| - \cdots + (-1)^{n+1} \left|\bigcap_{i=1}^{n} A_i\right|$$

Relations

Definition

A binary relation from A to B is any subset of

$$A \times B$$

Mathematically,

$$R \subseteq A \times B \; if \; (a, b) \in R$$

We write aRb meaning "a is related to b."

Example of a Binary Relation

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Let

$$A = \{1, 2, 3\}$$

Define

$$R = \{(1, 2), (2, 3), (1, 3)\}$$

Then

$$R \subseteq A \times A$$

Hence,

R is a binary relation on A.

A relation is often defined using a predicate.

Example

$$R = \{(a, b) \mid a, b \in A \land a < b\}$$

where

$$P(a, b) : a < b$$

is the predicate.

For

$$R \subseteq A \times B$$

the **Domain** is

$$Dom(R) = \{a \in A \mid \exists b \in B, (a, b) \in R\}$$

The **Range** is

$$Ran(R) = \{b \in B \mid \exists a \in A, (a, b) \in R\}$$

Properties of Relations

The six fundamental properties are

- Reflexive
- Irreflexive
- Symmetric
- Antisymmetric
- Asymmetric
- Transitive

These properties help determine whether a relation is an Equivalence Relation or a Partial Order.

Reflexive

$\forall a \in A,\ (a, a) \in R$

Every element is related to itself.

Example:

$\leq, \quad =, \quad \subseteq$

Irreflexive

$\forall a \in A,\ (a, a) \notin R$

No element is related to itself.

Example:

$<, \quad >$

Symmetric Relation

Definition

$$(a, b) \in R \implies (b, a) \in R$$

Example:

$$xRy \iff x \text{ and } y \text{ belong to the same class}$$

$$(Student_1, Student_2) \implies (Student_2, Student_1)$$

Definition

$$(a, b) \in R \wedge (b, a) \in R \implies a = b$$

Example:

$$\subseteq$$

If

$$A \subseteq B \text{ and } B \subseteq A$$

then

$$A = B$$

Definition

$$(a, b) \in R \implies (b, a) \notin R$$

Example:

$$<$$

If

$$3 < 5$$

then

$$5 < 3$$

is false.

Definition

$$(a, b) \in R \wedge (b, c) \in R \implies (a, c) \in R$$

Example:

$$\leq, \quad \subseteq$$

Software Dependency Graph

If

$$Module_A \to Module_B$$

and

$$Module_B \to Module_C$$

then

$$Module_A$$

indirectly depends on

$$Module_C$$

illustrating transitivity.

Equivalence Relations and Partial Orders

Definition

A relation R on a set A is called an **Equivalence Relation** if

$$R \text{ is } \boxed{\text{Reflexive}} \wedge \boxed{\text{Symmetric}} \wedge \boxed{\text{Transitive}}$$

Suppose an image classifier predicts

$$Images = \{Dog_1, Dog_2, Dog_3, Cat_1, Cat_2, Car_1\}$$

Define

$$xRy \iff x \text{ and } y \text{ belong to the same class}$$

Equivalence Classes

$$Dogs = \{Dog_1, Dog_2, Dog_3\}$$
$$Cats = \{Cat_1, Cat_2\}$$
$$Cars = \{Car_1\}$$

Partially Ordered Sets

Definition

A relation R on a set A is called a **Partial Order** if

$$R \text{ is } \boxed{\text{Reflexive}} \wedge \boxed{\text{Antisymmetric}} \wedge \boxed{\text{Transitive}}$$

A set together with a partial order,

$$(A, R)$$

is called a

$$\boxed{\textbf{Partially Ordered Set (Poset)}}$$

Let

$$A = \{1, 2, 3\}$$

Its power set is

$$\mathcal{P}(A) = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}$$

Define

$$R = \subseteq$$

Then

$$(\mathcal{P}(A), \subseteq)$$

is a Partially Ordered Set.

A software project contains the tasks

$$\textit{Collect} \rightarrow \textit{Design} \rightarrow \textit{Develop} \rightarrow \textit{Test} \rightarrow \textit{Deploy}$$

Define

$$A \leq B$$

to mean

$$\text{Task } A \text{ must be completed before Task } B.$$

This dependency relation is

- Reflexive
- Antisymmetric
- Transitive

Hence, it forms a Partial Order.

Hasse Diagrams

Definition

A **Hasse Diagram** is a graphical representation of a partially ordered set (poset), obtained by removing self-loops, arrowheads and transitive edges.

Purpose

It provides a simplified visualization of the ordering relationship between the elements of a poset.

Hasse Diagram of $(\mathcal{P}(A), \subseteq)$

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Let

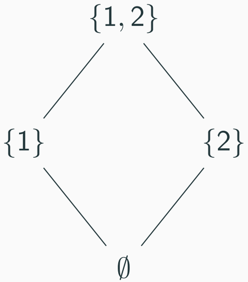
$$A = \{1, 2\}$$

Then

$$\mathcal{P}(A) = \{\emptyset, \{1\}, \{2\}, \{1, 2\}\}$$

with the partial order

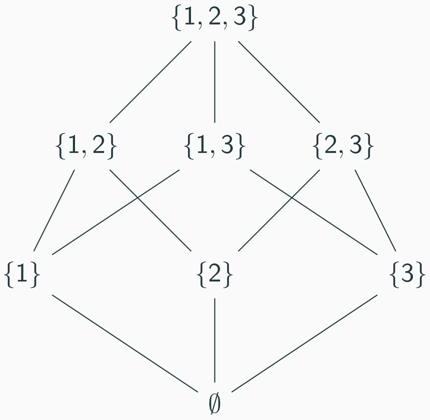
$$R = \subseteq .$$



Observation

- $\emptyset$ is the least element.
- $\{1, 2\}$ is the greatest element.
- $\{1\}$ and $\{2\}$ are incomparable.
- This is a partially ordered set.

Let $A = \{1, 2, 3\}, S = \mathcal{P}(A), R = \subseteq$. *Then,* (S, R) *is a partially ordered set*



Maximum, Minimum, Chain and Antichain

Let $(P, \leq)$ be a partially ordered set.

Maximum Element

An element $m \in P$ is called the **maximum (greatest)** element if

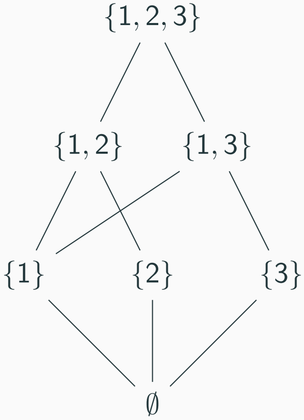
$$\forall x \in P,\ x \leq m.$$

Minimum Element

An element $n \in P$ is called the **minimum (least)** element if

$$\forall x \in P,\ n \leq x.$$

Example using a Hasse Diagram



$$\boxed{\begin{array}{l}\text{Minimum Element} = \emptyset \\ \text{Maximum Element} = \{1, 2, 3\}\end{array}}$$

Chain
A **chain** is a subset of a poset in which every pair of elements is comparable.

$$\forall a, b \in C,\ a \leq b \text{ or } b \leq a.$$

Example:

$$\emptyset \subseteq \{1\} \subseteq \{1, 2\} \subseteq \{1, 2, 3\}$$

Antichain
An **antichain** is a subset in which no two distinct elements are comparable.

$$\forall a \neq b,\ a \nleq b \wedge b \nleq a.$$

Example:

$$\{\{1\}, \{2\}, \{3\}\}$$

Thank You